B.Sc (HONS.) IN CSE FIRST YEAR, SECOND SEMESTER EXAMINATION, 2020

DIGITAL SYSTEMS DESIGN

[According to the New Syllabus]

Subject Code : 510221

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any **four** *questions. Different parts of a questions must be answered seqiuentially.]*

Marks

1. *(a)* Describe the main functional units of a digital computer. 5

   (b) Write short notes about the following terms: 4

   (i) BCD code

   (ii) Unicode — 16 bit

   (c) Represent the decimal number 396 in 6

   (*i*) Binary form

   (*ii*) Gray Code

   (*iii*) Excess-3 code

   (d) What are the steps actually required for processing an analog signal using digital hardware? Explain briefly. 5

2. *(a)* Implement the logic circuit that has the expression $X = \overline{AB\,(\overline{C + D})}$ using only NOR and NAND gates. 6

   (b) Differentiate between binary and BCD code. 3

   (c) What is Boolean algebra? Explain. the Boolean function $F = A+\overline{B}C$ in a sum of minterms. 6

   (d) Design a logic circuit with inputs L, M and N. So that the output P is high. Whenever the majority of inputs are high. 5

3. *(a)* What is setup and hold times? Explain J-k flipflop with suitable diagrams. 2+6=8

   (b) How can a J-k flipflop be modified to operate as a D flipflop and a T flipflop? 6

   (c) What is NAND gate latch? Explain clocked S-C flipflop with fiming diagram. 2+4=6

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4. (a) Explain the difference between a DEMUX and a MUX.

(b) Design and explain four-input multiplexer. 6

(c) What is semiconductor memory? Discuss the classification of semiconductor memory. 6

(d) Describe static and dynamic RAM. 4

5. (a) What is Decoder? Explain 3 to 8 line Decoder with truth table and circuit diagram. 6

(b) What is Karnaugh map? Using K-map simplify the following expression: 4

$$F(A, B, C, D) = \sum m(1,3,4,6,8,9,11,13,15) + d(0,2,14)$$

(c) Show how to wire a 74LS293 as a MOD-14 counter. 4

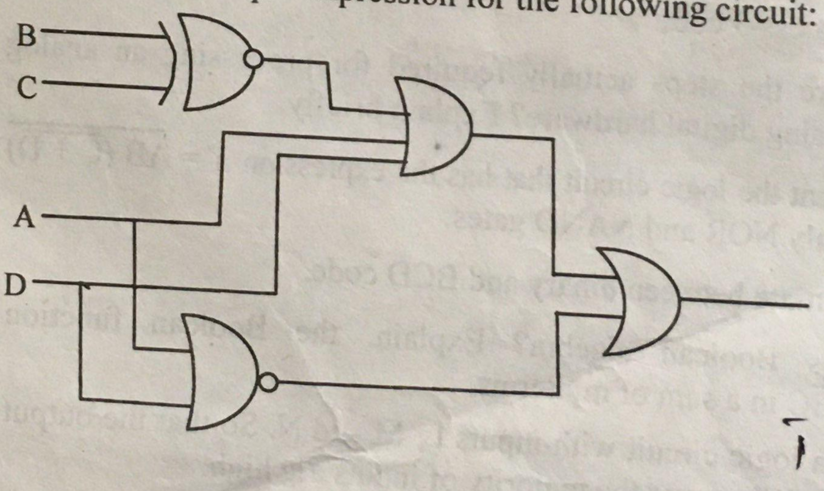
(d) Design a full adder circuit using basic logic gates and explain its operations with truth table. 6

6. (a) State the purpose of analog to digital conversion. 5

(b) Mention some applications of Decoder. 5

(c) Explain the operation of a 4 bit R-2R ladder DAC. 6

(d) Determine the output expression for the following circuit: 4



3 9 6 — 1001 1001 — 0110 = 0101

0011 1001 0010 1101

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